

Practice Quiz: Planning

Part A: Multiple Choice

1. In Active Inference, planning is formalized as: A) A search through a decision tree using minimax B) Policy selection — choosing the action sequence that minimizes Expected Free Energy C) Random action followed by reinforcement learning D) Conscious deliberation followed by voluntary choice
2. Expected Free Energy (G) for a policy π decomposes into: A) Perception error and action error B) Risk (divergence from preferred outcomes) and ambiguity (expected uncertainty) C) Complexity and accuracy D) Past evidence and future predictions
3. Sophisticated inference extends basic policy selection by: A) Including emotional states in the generative model B) Considering what the agent will *believe* after each action (counterfactual belief updating) C) Using faster computational hardware D) Eliminating uncertainty entirely before acting
4. Husserl's concept of "protention" refers to: A) The memory of past events B) The anticipation of what is about to happen in the near future C) The uncertainty about distant future events D) The desire to protect oneself from harm
5. Aristotle's concept of *telos* relates to Active Inference through: A) The material composition of the generative model B) The prior preferences (C vector) that bias policy selection toward preferred outcomes C) The efficient cause of neural activity D) The formal structure of Bayesian inference
6. Active Inference explains procrastination as: A) Laziness — a deficit in motivation B) A flat EFE landscape where the agent cannot distinguish between policies C) An excess of planning depth D) A failure of motor control
7. If prior preferences (C vector) are products of evolution and can be modified by learning, then: A) Purpose is fixed at birth and cannot change B) The agent's purposes

are both inherited and self-constructed to some degree C) Only biological organisms can have purposes D) Artificial agents can never plan

Part B: Short Answer

1. Explain how sophisticated inference involves "reasoning about what you will believe" after acting. Give a concrete example.
2. Compare the Active Inference account of planning with Heidegger's concept of *Dasein* as "being-toward" the future. What philosophical insight does this comparison reveal?
3. L.A. Paul's concept of "transformative experience" suggests that some choices fundamentally change the chooser. Analyze this concept using the Active Inference framework: can an agent rationally plan for a transformative experience?